

Python for Data Science - 2305CS303

Lab - 8

Roll No. : 128

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01) Write a function to calculate BMI given mass and height.
($BMI = mass/h^{**2}$)

```
In [5]: def bmi(m, h):  
        return m / h**2  
  
        m = float(input("Enter mass: "))  
        h = float(input("Enter height: "))  
  
        print("BMI =", bmi(m, h))
```

BMI = 26.666666666666668

02) Write a function that add first n numbers.

```
In [4]: def add(n):  
        s = 0  
        for i in range(1, n + 1):  
            s = s + i  
        return s  
  
        n = int(input("Enter n: "))  
        print(add(n))
```

6

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

```
In [2]: def prime(n):  
        if n < 2:  
            return 0
```

```

    for i in range(2, n):
        if n % i == 0:
            return 0

    return 1

n = int(input("Enter number: "))
print(prime(n))

```

1

04) Write a function that returns the list of Prime numbers between given two numbers.

```

In [1]: def primes(a, b):
        l = []

        for n in range(a, b + 1):
            if n < 2:
                continue

            for i in range(2, n):
                if n % i == 0:
                    break
            else:
                l.append(n)

        return l

print(primes(1,10))

```

[2, 3, 5, 7]

05) Write a function that returns True if the given string is Palindrome or False otherwise.

```

In [7]: def pal(s):
        if s == s[::-1]:
            return True
        else:
            return False

s = input("Enter string: ")
print(pal(s))

```

False

06) Write a function that returns the sum of all the elements of the list.

```

In [8]: def add(l):
        s = 0
        for i in l:
            s = s + i
        return s

```

```
l = [10, 20, 30, 40]
print(add(l))
```

100

07) Write a function to calculate the sum of the first element of each tuples inside the list.

```
In [9]: def add(l):
        s = 0
        for i in l:
            s = s + i[0]
        return s

l = [(10, 20), (30, 40), (50, 60)]

print(add(l))
```

90

08) Write a recursive function to find nth term of Fibonacci Series.

```
In [12]: def fib(n):
        if n == 0:
            return 0
        if n == 1:
            return 1
        return fib(n - 1) + fib(n - 2)

n = int(input("Enter n: "))
print(fib(n))
```

8

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

```
In [13]: def name(r):
        d = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'}
        return d[r]

r = int(input("Enter rno "))
print(name(r))
```

Rahul

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

```
In [ ]: def add(l):
        a = []

        for i in l:
            if str(i).endswith("0"):
                a.append(i)

        return sum(a)

l = [200, 456, 300, 100, 234, 678]

print(add(l))
```

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

```
In [14]: def invert(d):
        n = {}

        for k, v in d.items():
            n[v] = k

        return n

d = {'a': 10, 'b': 20, 'c': 30, 'd': 40}

print(invert(d))
```

```
{10: 'a', 20: 'b', 30: 'c', 40: 'd'}
```

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Oupptput : no_upper = 3, no_lower = 5

```
In [15]: def count(s):
        u = 0
        l = 0

        for i in s:
            if i.isupper():
                u += 1
            if i.islower():
                l += 1

        print("no_upper =", u)
        print("no_lower =", l)

s = "AbcDEfgh"
count(s)
```

```
no_upper = 3
no_lower = 5
```

13) Write a lambda function to get smallest number from the given two numbers.

```
In [16]: small = lambda a, b: min(a, b)

print(small(10, 5))
```

5

14) For the given list of names of students, extract the names having more than 7 characters. Use filter().

```
In [17]: l = ['Ajay', 'Rahul', 'Raj', 'Abhishek', 'Pyl']

a = list(filter(lambda x: len(x) > 7, l))

print(a)
```

```
['Abhishek', 'Pratiksha']
```

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

```
In [18]: l = ['ajay', 'rahul', 'jay', 'pyl']

a = list(map(lambda x: x.capitalize(), l))

print(a)
```

```
['Ajay', 'Rahul', 'Jay', 'Pyl']
```

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments
3. Default Arguments
4. Variable Length Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

```
In [19]: def add(a, b=10):
          print(a + b)

          add(20)

          def add(a, b):
              print(a + b)

          add(a=10, b=20)

          def add(a, b=10):
```

```
    print(a + b)

add(20)

def add(*args):
    print(sum(args))

add(10, 20, 30, 40)

def add(a, b, /):
    print(a + b)

add(10, 20)

def add(*, a, b):
    print(a + b)

add(a=10, b=20)
```

```
30
30
30
100
30
30
```

In []: